

Documentation of Git Commands

1. git init

Purpose: Initializes a new Git repository in your project folder.

Creates a .git folder — now Git will track changes in this folder.

2. git clone <repository-url>

Purpose: Copies a remote Git repository (like from GitHub) to your local machine.

Creates a local copy of the entire repository.

3. git status

Purpose: Shows the current status of your working directory and staging area.

Tells you which files are:

- Modified
- Staged
- Untracked

4. git add <filename> or git add .

Purpose: Adds file(s) to the staging area.

Files are now staged and ready for commit.

5. git commit -m "message"

Purpose: Records the staged changes with a message
Saves the snapshot of changes in your project history.

6. git log

Purpose: Displays a history of commits.

Shows:

- Commit hash
- Author
- Date
- Commit message

7. git diff

Purpose: Shows changes between working directory and staging or last commit.

See what you've changed before committing.

8. git branch

Purpose: Lists all branches or creates a new one.

Helps in parallel development.

9. git switch <branch> or git checkout <branch>

Purpose: Switches between branches.

Changes your working directory to that branch.

10. git merge <branch>

Purpose: Merges the changes from another branch into the current branch.

Useful when combining features.

11. git pull

Purpose: Fetches and merges changes from the remote repository.

Keeps your local repo up to date.

12. git push

Purpose: Uploads local commits to the remote repository.

Shares your changes with others.

13. git remote -v

Purpose: Shows the URLs of remote repositories.

14. git rm <filename>

Purpose: Removes a file from the staging area and working directory.

15. git reset <filename> or git reset --hard

Purpose: Unstages files or resets your code to a previous state.

Warning: --hard deletes changes permanently.

16. git config

Purpose: Configures Git settings like username and email.

git config --global user.name "Your Name"

git config --global user.email "you@example.com"

`git config --list` # Shows all config

Essential before making commits.

17. git stash

Purpose: Temporarily saves uncommitted changes.

`git stash` # Save uncommitted changes

`git stash pop` # Reapply the changes

`git stash list` # View stashes

Useful when switching branches without committing.

18. git fetch

Purpose: Gets updates from remote without merging.

Use `git fetch` + `git log` or `git diff` to review changes before applying.

19. git show <commit-hash>

Purpose: Displays details of a specific commit.

Shows commit message, changes, author, date.

20. git revert <commit-hash>

Purpose: Undoes a commit **without deleting history**.

Safe way to undo a change in public repositories.

21. git tag

Purpose: Marks a specific point in history (commonly for releases).

`git tag v1.0`

```
git tag -a v1.1 -m "Version 1.1"
```

```
git push origin v1.0
```

Helps in versioning.

22. git clean

Purpose: Removes untracked files.

```
git clean -n    # Preview
```

```
git clean -f    # Remove untracked files
```

Use carefully — it can delete your files. **23. git blame <file>**

Purpose: Shows who made which changes in each line of a file.

Great for debugging or tracking history.

java program to print Hello

```
atiwa@HP MINGW64 /e/Git Java Code
$ mkdir HelloJavaProject

atiwa@HP MINGW64 /e/Git Java Code
$ cd HelloJavaProject

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject
$ touch Hello.java

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject
$ nano Hello.java

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject
$ java -version
java version "1.8.0_202"
Java(TM) SE Runtime Environment (build 1.8.0_202-b08)
Java HotSpot(TM) 64-Bit Server VM (build 25.202-b08, mixed mode)

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject
$ ls
Hello.java

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject
$ javac Hello.java

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject
$ java Hello
Hello

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject
$ git init
Initialized empty Git repository in E:/Git Java Code/HelloJavaProject/.git/

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject (master)
$ git add Hello.java
warning: in the working copy of 'Hello.java', LF will be replaced by CRLF the next time Git touches it

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject (master)
$ git commit -m "Add Hello.java to print Hello"
[master (root-commit) fb70749] Add Hello.java to print Hello
1 file changed, 5 insertions(+)
create mode 100644 Hello.java

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject (master)
$ git remote add origin https://github.com/Adityatiwari001/webtech

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject (master)
$ git branch -m master main

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject (main)
$ git push -u origin main
fatal: User cancelled dialog.
remote: No anonymous write access.
fatal: Authentication failed for 'https://github.com/Adityatiwari001/webtech/'

atiwa@HP MINGW64 /e/Git Java Code/HelloJavaProject (main)
$
```

Basic commands

```
atiwa@HP MINGW64 /e/Git Home Work
$ git init
Initialized empty Git repository in E:/Git Home Work/.git/

atiwa@HP MINGW64 /e/Git Home Work (master)
$ touch file.txt

atiwa@HP MINGW64 /e/Git Home Work (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    file.txt

nothing added to commit but untracked files present (use "git add" to track)

atiwa@HP MINGW64 /e/Git Home Work (master)
$ git add file1.txt
fatal: pathspec 'file1.txt' did not match any files

atiwa@HP MINGW64 /e/Git Home Work (master)
$ git commit -m "Hello"
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    file.txt

nothing added to commit but untracked files present (use "git add" to track)

atiwa@HP MINGW64 /e/Git Home Work (master)
$ git log
fatal: your current branch 'master' does not have any commits yet

atiwa@HP MINGW64 /e/Git Home Work (master)
$ git diff

atiwa@HP MINGW64 /e/Git Home Work (master)
$ git branch
```

```
atiwa@HP MINGW64 /e/Git Home Work (master)  
$ git branch -m master main
```

```
atiwa@HP MINGW64 /e/Git Home Work (main)  
$ git branch value-x  
fatal: not a valid object name: 'main'
```

```
atiwa@HP MINGW64 /e/Git Home Work (main)  
$ git switch -c values  
Switched to a new branch 'values'
```

```
atiwa@HP MINGW64 /e/Git Home Work (values)  
$ git checkout -b features  
Switched to a new branch 'features'
```



```

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git merge values
merge: values - not something we can merge

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git pull origin main
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'origin'

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git remot add origin https://github.com/Adityatiwari001/webtech
git: 'remot' is not a git command. See 'git --help'.

The most similar command is
    remote

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git remote add origin https://github.com/Adityatiwari001/webtech

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git merge values
merge: values - not something we can merge

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git pull origin main
remote: Enumerating objects: 110, done.
remote: Total 110 (delta 0), reused 0 (delta 0), pack-reused 110 (from 1)
Receiving objects: 100% (110/110), 10.32 MiB | 2.09 MiB/s, done.
Resolving deltas: 100% (7/7), done.
From https://github.com/Adityatiwari001/webtech
 * branch          main       -> FETCH_HEAD
 * [new branch]    main       -> origin/main

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Adityatiwari001/webtech'

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git push origin features
fatal: User cancelled dialog.
error: unable to read askpass response from 'C:/Program Files/Git/mingw64/bin/git-askpass.exe'
Password for 'https://github.com':
remote: No anonymous write access.
fatal: Authentication failed for 'https://github.com/Adityatiwari001/webtech/'

atiwa@HP MINGW64 /e/Git Home Work (features)
$ |

```

```
atiwa@HP MINGW64 /e/Git Home Work (features)
$ git stash
No local changes to save

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git stash pop
No stash entries found.

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git stash list

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git fetch origin

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git show 1234abcd
fatal: ambiguous argument '1234abcd': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git clean -n
Would remove file.txt

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git clean -f
Removing file.txt

atiwa@HP MINGW64 /e/Git Home Work (features)
$ git blame main.py
fatal: no such path 'main.py' in HEAD

atiwa@HP MINGW64 /e/Git Home Work (features)
$ |
```